U O W

CSIT305 Emerging Information Technologies and their Applications

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Subject Coordinator

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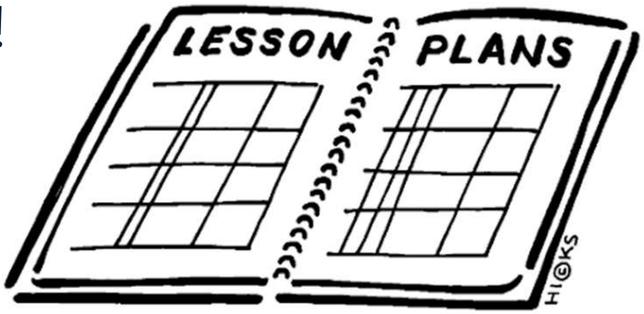
  - https://scholars.uow.edu.au/display/jun_yan

CSIT305 Things we need to know

Let’s go through the Subject Outline!



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Subject Description

- Advanced theories, algorithms and applications in the emerging information technologies
- Important concepts and principles for each topic covered
- Advanced approaches and methods

- This subject will equip students with advanced knowledge of emerging information technologies and enhance their skills to appropriately choose and apply information techniques to resolve practical problems effectively.

Subject Learning Outcomes

- On successful completion of this subject, students will be able to:
  - Identify and describe key concepts associated with emerging information technologies.
  - Explain the fundamental principles and potential applications of emerging information technologies in various domains, demonstrating an understanding of how these technologies can impact business processes and societal trends.
  - Apply basic emerging technologies to solve real-world problems by using tools and platforms relevant to the technologies studied.
  - Analyse the advantages and limitations of different emerging technologies and assess their suitability for specific business or research scenarios, considering factors such as scalability, cost, and implementation challenges.
  - Develop a comprehensive strategy for integrating multiple emerging technologies into a cohesive solution, and evaluate the potential impacts on organisational efficiency, security, and innovation.

Subject Outline

- It is a ‘contract’ between you and the University
- It tells you what you are required to do and what you are not allowed to do: e.g.
  - It tells you when assignments are due, how and when they are to be submitted etc.
  - It tells you about penalties for late submission, for cheating and for plagiarism
  - It tells you about the textbook and other readings you should complete
- Your subject outline is available online at **CSIT305 Moodle site**

Moodle - eLearning

- This subject will use a Moodle (eLearning) site
  - Make sure you can log on
- Subject outline is/will be uploaded on Moodle.
  - Read through carefully
- Announcements posted to the eLearning site will be **deemed to have been delivered to you**
- Lecture slides (in PDF form) will become available there

Subject Materials

- **Recommended Readings**
  - Ahmed, W 2024, Advanced Database Systems 1st ed, Toronto Academic Press, Burlington.
  - Sakr, M, Vaisman, A & Zimányi, E 2025, Mobility Data Science: From Data to Insights 1st edn, Springer, Cham.
  - Shao, Y, Cui, B & Chen, L 2020, Large-scale Graph Analysis: System, Algorithm and Optimization 1st ed. 2020., Springer Singapore, Singapore.
  - Jo, T 2024, Text Mining: Concepts, Implementation, and Big Data Challenge Second Edition 2024., Springer Nature Switzerland, Cham.
  - Chung, J-M 2024, Emerging Secure Networks, Blockchains and Smart Contract Technologies First edition., Springer, Cham, Switzerland.
  - Romero, JR, Medina-Bulo, I & Chicano, F 2023, Optimising the Software Development Process with Artificial Intelligence 1st ed. 2023., Springer Nature Singapore, Singapore.
  - Ramachandran, Muthu & Mahmood, Zaigham (eds) 2020, Software Engineering in the Era of Cloud Computing 1st ed. 2020., Springer International Publishing, Cham.
  - Pal, S, Savaglio, C, Minerva, R & Delicato, FC 2024, IoT Edge Intelligence 1st ed. 2024., Springer Nature Switzerland, Cham.

- Students are encouraged to use the UOW Library catalogue and databases to locate additional resources including the e-readings list: https://ereadingsprd.uow.edu.au/

Subject Organisation

- Three-hour lecture
  - Focus on essential topics in lectures
- Three-hour lab
  - Lab exercises
  - Lab discussions
- Fuel your self-learning after classes

Subject Topic Sequence

(1) Advanced Database Overview
(2) Spatial Data and Computing
(3) Graph Data and Computing
(4) Text Data and Processing

Multi-Modal Database

(1) AI4SE: artificial intelligence for software engineering
(2) Software Engineering in Cloud Computing I
(3) Software Engineering in Cloud Computing II

Software

(1) IoT and Edge Computing
(2) Digital Twin Technology
(3) Blockchains and Smart Contracts
(4) Emerging Cyber-Attacks

Networking

Subject Assessments

ASSESSMENT ITEMS & FORMAT	% OF FINAL MARK	GROUP/ INDIVIDUAL	DUE DATE
Assignments	20%	Individual	Full-time students Quiz 1: T3 Quiz 2: T6 Part-time students Quiz 1: T2 Quiz 2: T4
Group Project	25%	Group	25th August
Presentation	5%	Group	Last tutorial
Final Examination	50%	Individual	In Examination period

To be eligible for a Pass in this subject a student must achieve a mark of at least 40% in the Final Examination. Students who fail to achieve this minimum mark may be given a TF (Technical Fail) for this subject.

Note on Subject Assessment

- Assessment guide is available on the subject's Moodle site
  - Instructions
  - Marking criteria

- All individual assignments must be completed independently

- Plagiarism in any assignment or exam may result in a FAIL for that assignment or exam

Note on Subject Assessment

- Penalties apply to all late assessments, except if student academic consideration has been granted. A new submission date may be given if Student Academic Consideration has been granted, however the late penalties below apply if not received by the new date.
- Late assignment submissions will attract a penalty of **5% of the total possible marks of the assessment item for each 24-hour period** or part thereof that the item is late, to a maximum penalty of receiving zero marks for the assessment item.
- Submissions received **4 days after the due date** will not be marked and will receive a mark of 0.
- If an assessment is submitted late, it will be marked in the normal way, and a penalty will then be applied.

Note on Subject Assessment – About GenAI

- **Generative AI is not a substitute for decision-making**: GenAI should complement, not replace, your critical thinking and decision-making skills.
- **Transparency in use**: Where required, you must acknowledge GenAI use, including providing prompt histories and detailing how GenAI was utilised.
- **Protect data and copyright**: Many GenAI technologies collect information in ways that breach privacy and data protection provisions, particularly where the source material is confidential or subject to copyright.

Technical Fail

- To be eligible for a Pass in this subject a student must achieve a mark of **at least 40%** in the Final Examination.

- Students who fail to achieve this minimum mark & would have otherwise passed may be given a TF (Technical Fail) for this subject.

Introduction to Emerging Information Technologies

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Outline

- Emerging Information Technologies
- Subject Plan

Breakthroughs that change our lives

- For a technology to be considered disruptive it has to modify a habit or behaviour and be accessible to the majority of the population.

- Examples of disruptive technologies [1]:
  - Blockchain, 5G technology, Advanced Virtual Reality, Artificial Intelligence and Machine Learning, Cloud services, Nanotechnology, Big Data, 3D printing, Cybersecurity, Quantum computing, Hyper-personalization, Chatbots and smart assistants, Smart Cities, Computational chemistry, Fingerprints

[1] https://www.repsol.com/en/energy-and-the-future/technology-and-innovation/disruptive-technologies/index.cshtml

Question

- Please watch the following video to list some emerging information technologies that transform IT industry and think about their applications in our daily life.

- https://www.youtube.com/watch?v=W6ghobNmAG8

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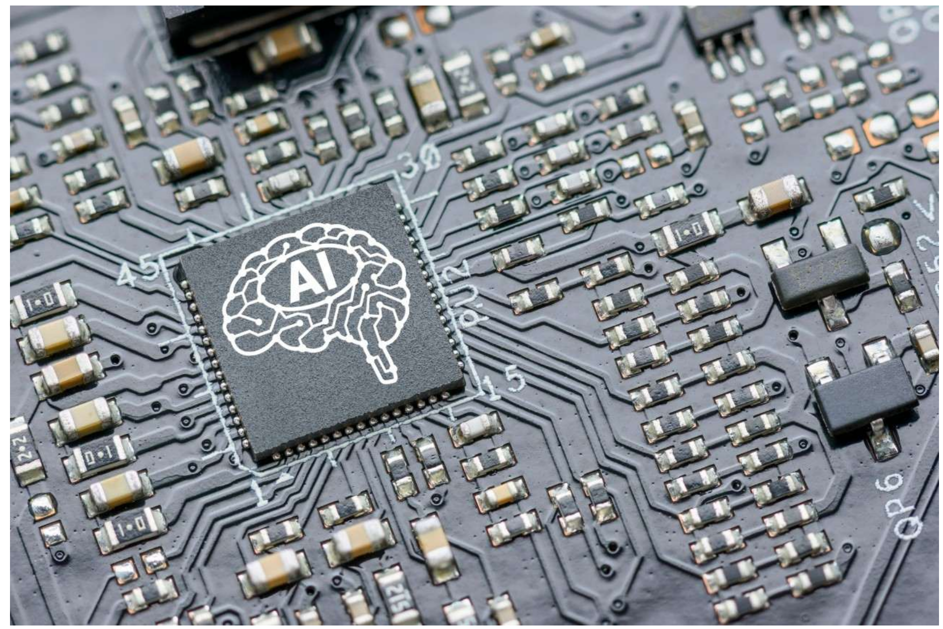
Big Data

- Big Data collects and manages large volumes of data that are then analyzed and applied to a specific area. We can say that it is a smart analysis tool to make more profitable and efficient decisions.
- It represents a great competitive advantage for companies because it reduces costs, saves time, decreases margin for error in processes, and improves the quality of the final product.

Artificial Intelligence and Machine Learning

- Artificial Intelligence (AI) is the combination of algorithms that are programmed so that a machine carries out actions that until now were carried out by humans, such as learning from data analysis or planning.
- Machine Learning is a discipline within the AI field that allows computers to learn for themselves and carry out tasks autonomously.



Cloud Services

- Cloud services, also known as cloud computing, allow files to be saved on the internet without needing external storage equipment.
- It represents great advantages for companies because it allows users to enjoy management tools from any place in the world by just connecting from their device.

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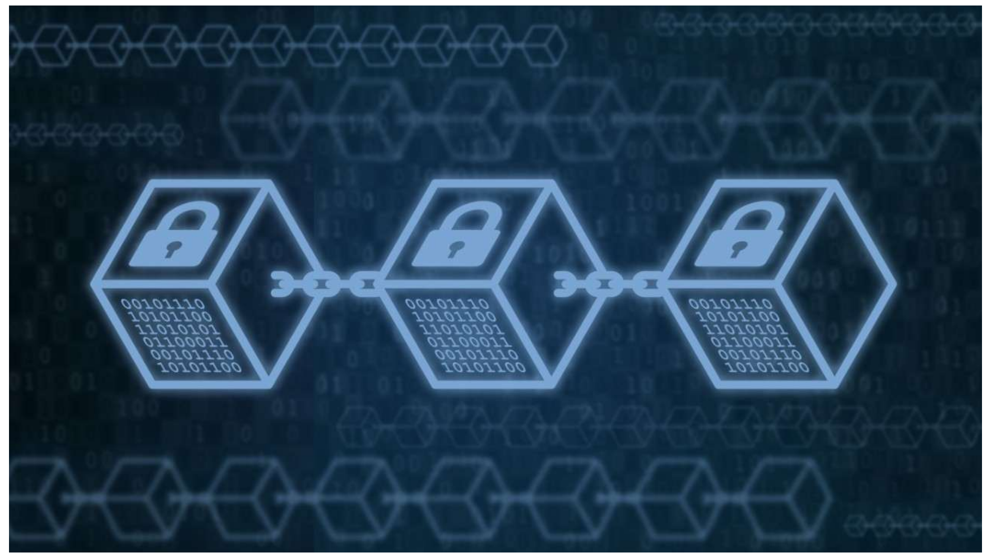
Blockchain

- Blockchain technology, like tokeni zation is applied in a digitalized workflow to create shared, transparent, and more secure processes.
- It works like a chain of blocks in which all nodes are connected to each other, so it guarantees the traceability of any process, creating a unique registration network that reduces risks and costs.



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5G Technology

- What does incorporating 5G technology imply?
  - It means a change in the telecommunications sector. 5G will definitively introduce the Internet of Things (IoT) into our daily lives.
  - This disruptive technology will make it possible for all devices to be connected and share information at high speed.

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Cybersecurity

- Cybersecurity or technology security is the set of processes and tools that are implemented to save information that any device, program, or company generates. With the use of network systems, software, and applications, the execution of security protocols is necessary to protect confidential, private, or sensitive information.

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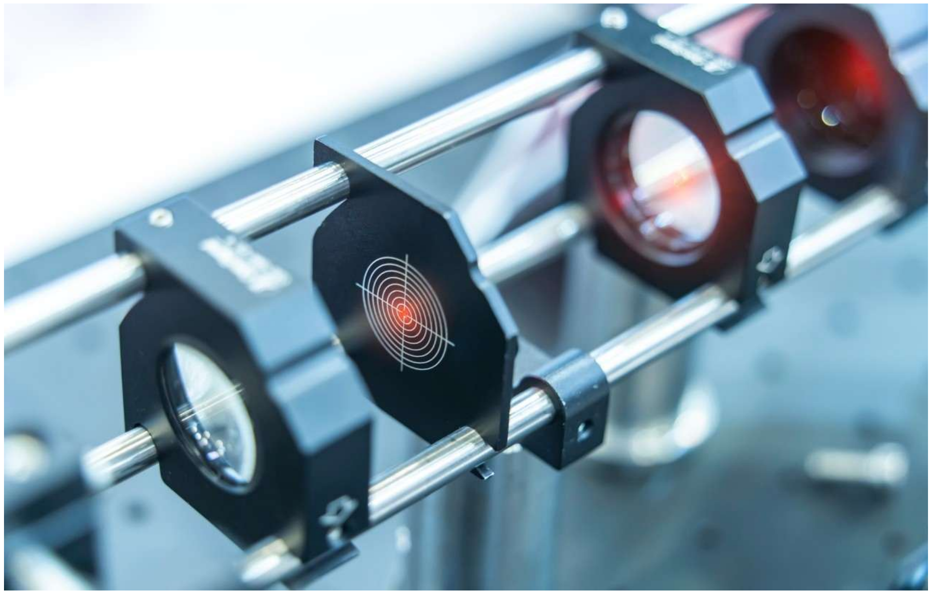
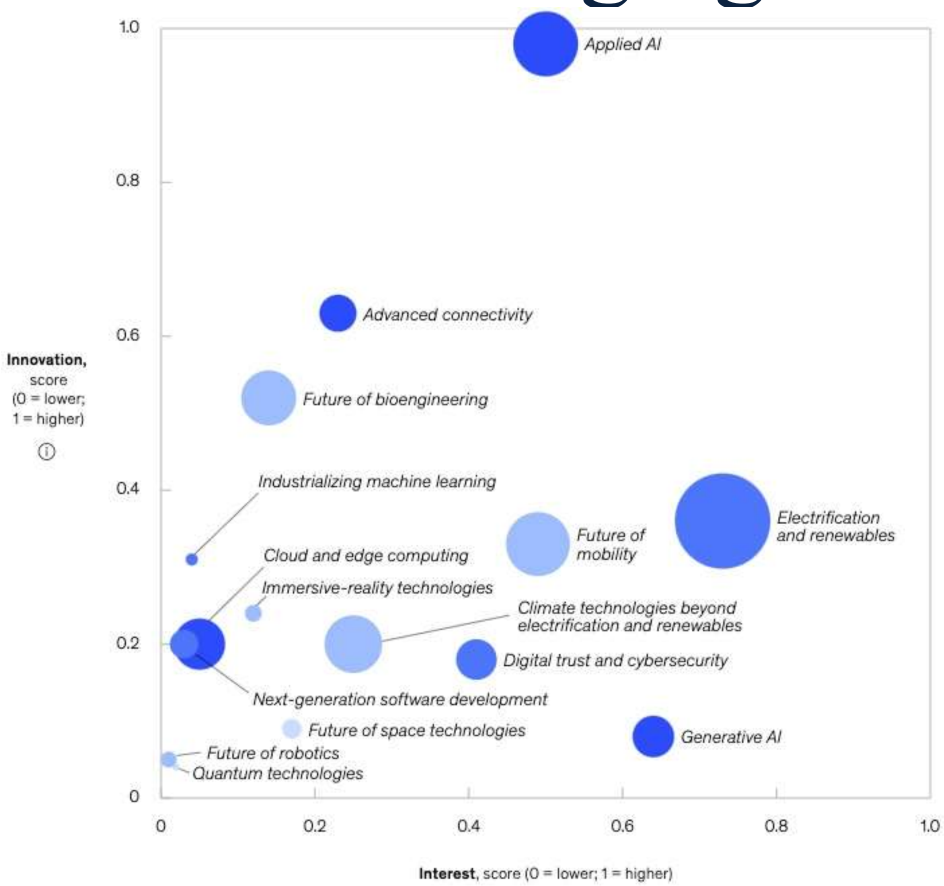
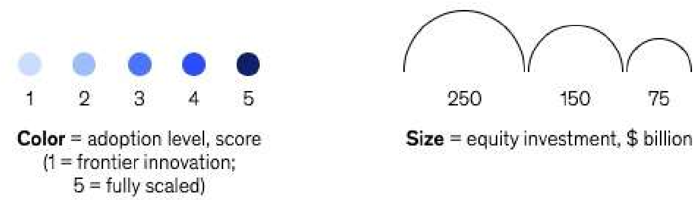
Quantum Computing

- Quantum computing contributes to solving complex problems and creating effective algorithms in little time.
- This paradigm allows information to be processed at very high levels with multiple applications.
  - For example, in the area of health, it could contribute to creating new personalized pharmaceuticals.
  - And in the finance sector, it could accurately anticipate changes in the market.



List of Emerging Information Technologies





- Innovation: based on patents and research
- Interest: based on news and web searches

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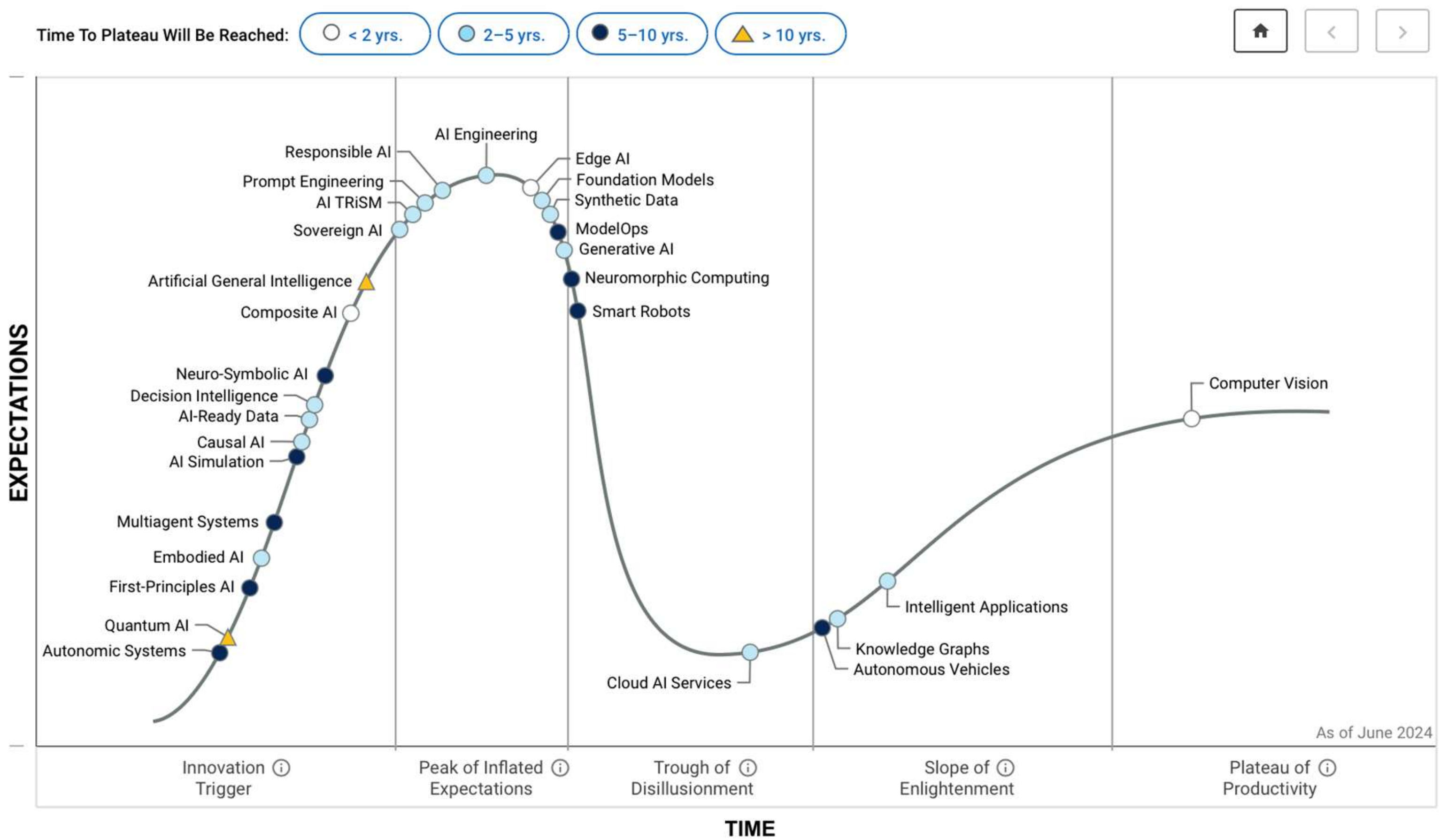
McKinsey Technology Trends Outlook 2024 published on July 16, 2024. The link: https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-top-trends-in-tech#new-and-notable

List of Emerging Information Technologies



[Gartner Hype Cycle. https://www.gartner.com/

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List of Emerging Information Technologies

How does a Gartner Hype Cycle™ work?

A Gartner Hype Cycle provides an objective map that helps you understand the real risks and opportunities of innovation, so you can avoid adopting something too early, giving up too soon, adopting too late, or hanging on too long. Every Hype Cycle includes five phases:

The innovation trigger starts when an event, like a technological breakthrough or a product launch, gets people talking.

The peak of inflated expectations is when product usage increases, but there's still more hype than proof that the innovation can deliver what you need.

The trough of disillusionment happens when the original excitement wears off and early adopters report performance issues and low ROI.

The slope of enlightenment occurs when early adopters see initial benefits and others start to understand how to adapt the innovation to their organizations.

The plateau of productivity marks the point at which more users see real-world benefits and the innovation goes mainstream.

[1] https://www.gartner.com/en/documents/5569027

The New Jobs of the Future

- These are the 12 professions you may consider to further study
  - New energy engineer
  - Smart Cities and smart homes architect and engineer
  - Digital transformation specialist
  - Climate defender: Carbon footprint controller
  - Big Data analyst
  - Biotechnology engineer
  - IoT specialist
  - Cybersecurity expert
  - New materials engineer
  - Fintech engineer
  - Expert in Artificial Intelligence and machine learning
  - Smart transportation engineer

https://www.repsol.com/en/energy-and-the-future/people/professions-of-the-future/index.cshtml

Question

- Example: Smart Healthcare Monitoring System
  - A hospital wants to develop a Smart Healthcare Monitoring System to continuously track patient health through various devices.
  - What are the key functions can you think about in the system?

Key Functions Example

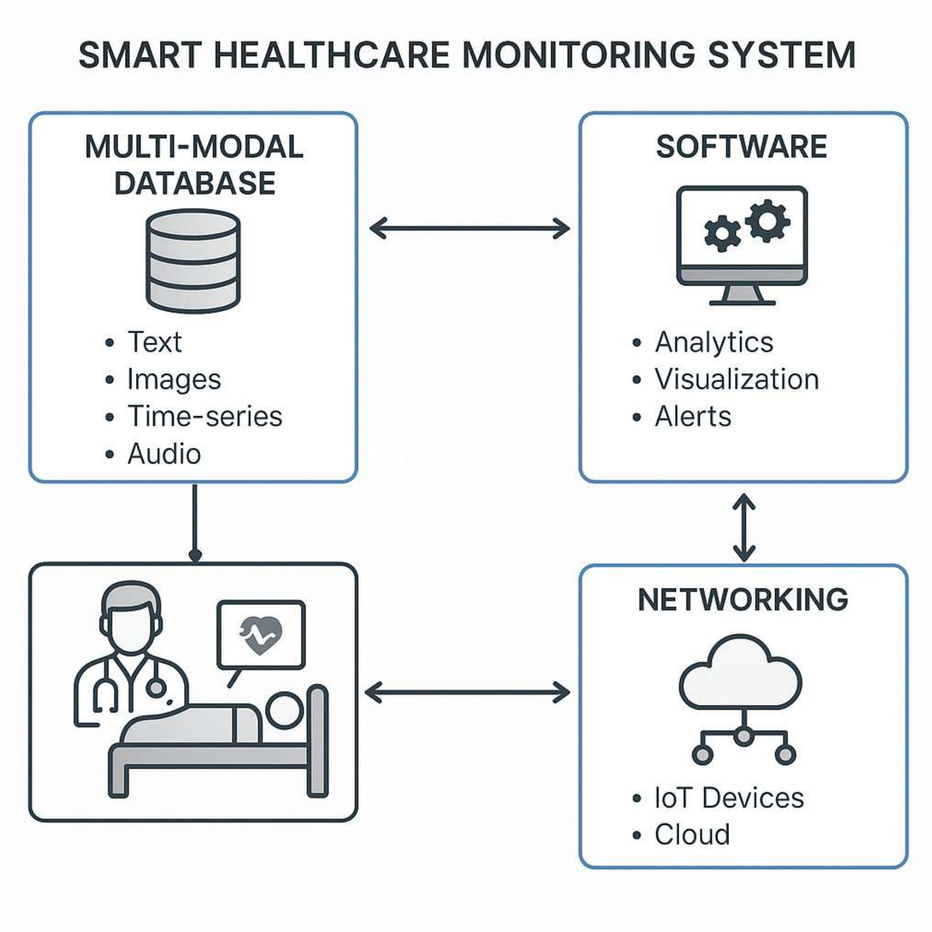
- Patient Monitoring
  - Continuously tracks vital signs (heart rate, etc.) and Alerts staff in case of anomalies
  - Supports remote and in-hospital patient observation

- Data Storage
  - Collects multi-modal data
    - Text: medical records, clinical notes
    - Images: X-rays, CT, MRI
    - Time-series: ECG, glucose levels
    - Audio: voice notes or patient interactions
  - Stores all data in a unified, searchable multi-modal database

Key Functions Example

- Intelligent Analysis
  - Trend analysis for chronic disease management
  - Predictive analytics for emergency prevention
- System Interoperability
  - Integrates with Electronic Health Records (EHR) systems
  - Interfaces with third-party diagnostic tools and apps
- Secure Networking & Communication
  - Transmits data securely across devices (wearables and medical devices) and locations
  - Enables telemedicine and cross-hospital consultations
  - Supports mobile access by authorized staff

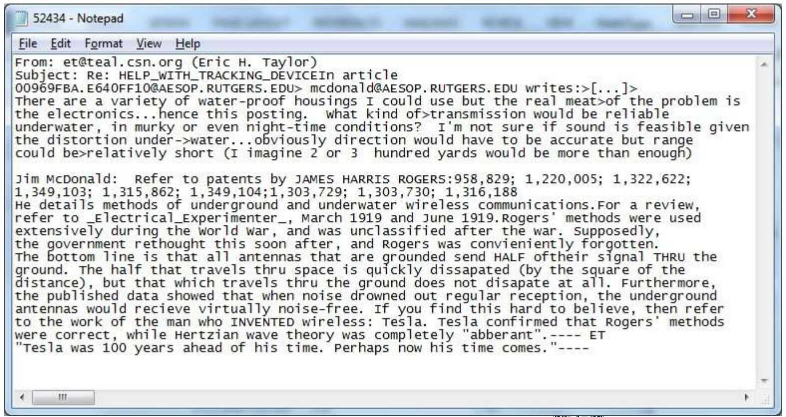
The Connections Among Three Modules



- The **multi-modal database** stores the rich variety of data and supports advanced querying and integration.
- The **network** transfers real-time data from IoT devices to servers.
- The **software** processes and analyzes this data to provide insights.
- Together, they form an intelligent, responsive healthcare system that enhances patient care.

Lecture Schedule

- Module 1 - Multi-Modal Database
  - Unstructured dataset
    - Spatial data
    - Graph data
    - Text data



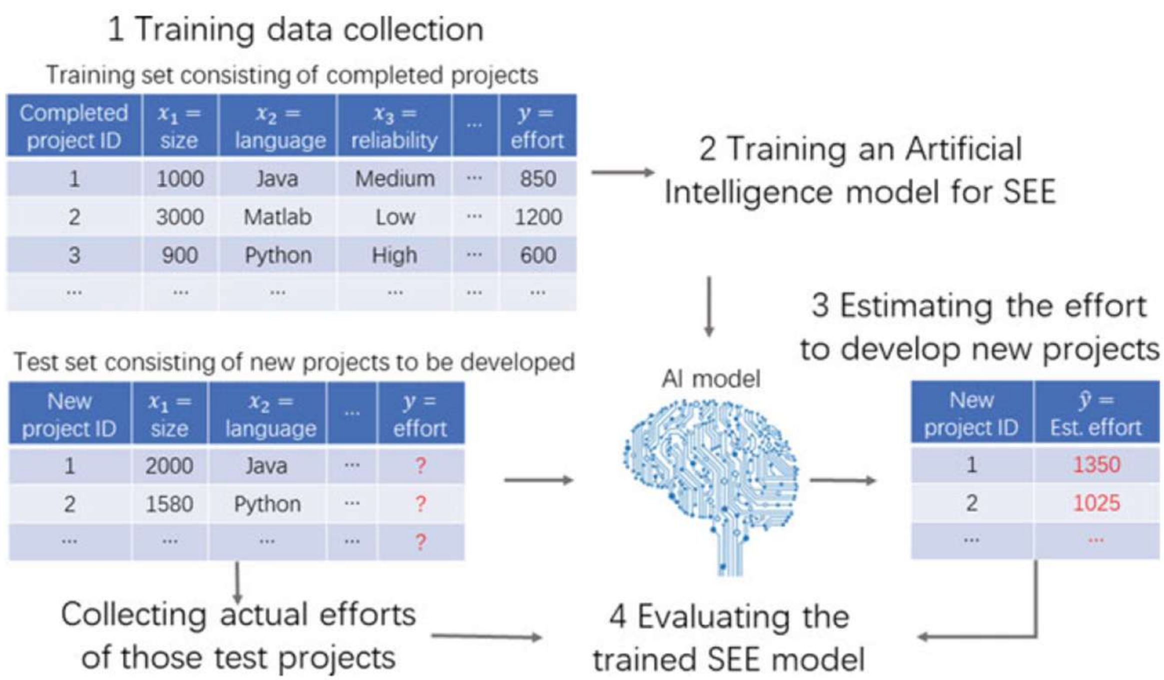
Example

- For example, the system collects data from multiple sources:
  - Text: Patient records, doctor notes.
  - Images: MRI scans, X-rays.
  - Time-series: Heart rate, blood pressure sensors.
  - Audio: Voice inputs for dictating symptoms or using virtual assistants.

- All this diverse data is stored in a **multi-modal database**, which supports querying and analyzing different data types in a unified manner.

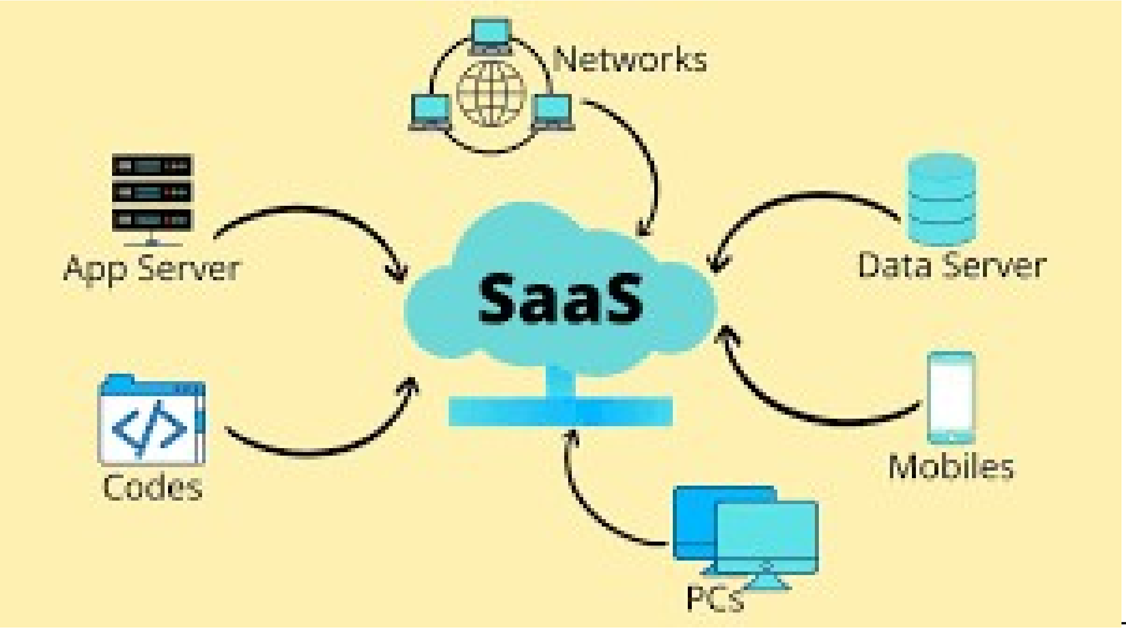
Lecture Schedule

- Module 2 – Software
  - Artificial Intelligence for Software Engineering



Lecture Schedule

- Module 2 – Software
  - Software engineering in cloud computing



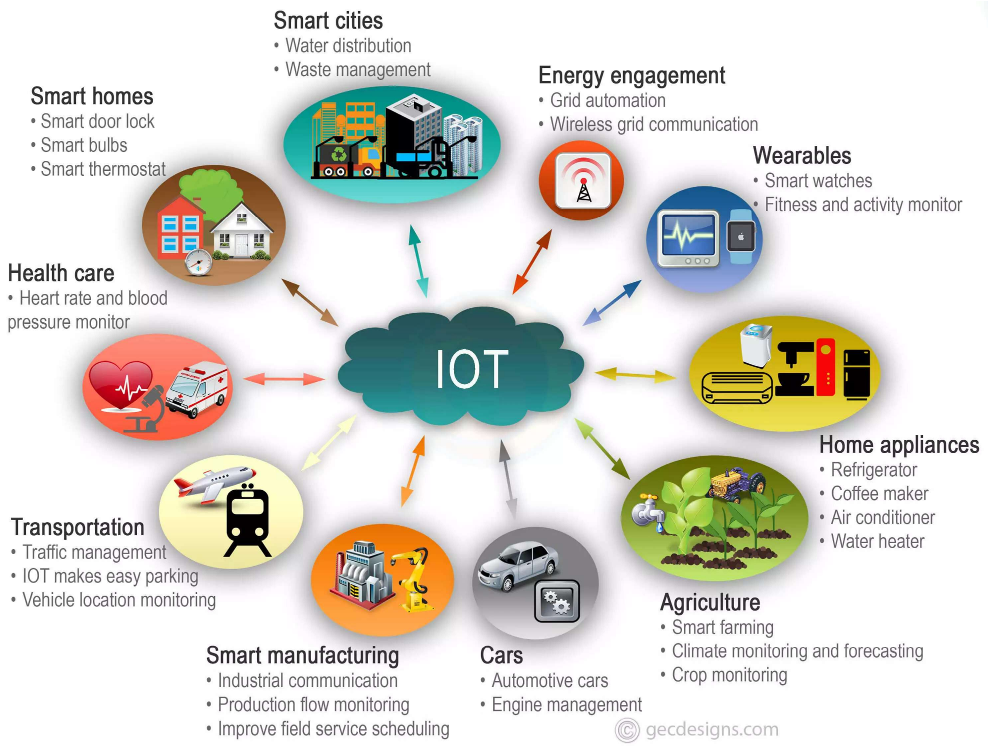
Example

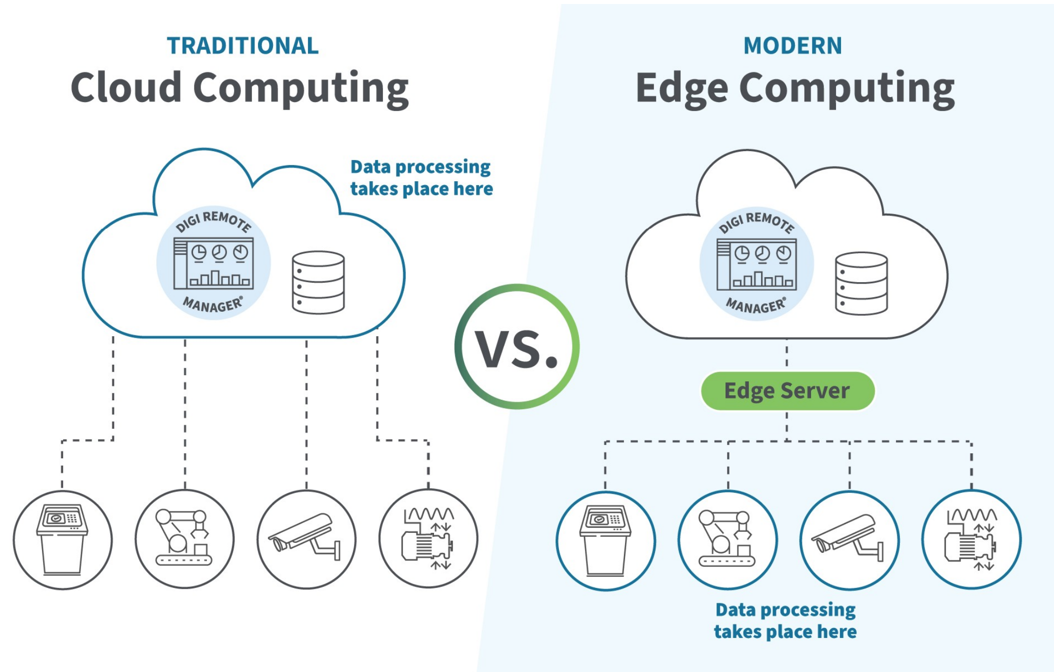
- For example, Custom healthcare software processes the data:
  - Analyzes sensor data using machine learning.
  - Detects abnormal patterns (e.g., irregular heartbeats).
  - Displays real-time dashboards to doctors.
  - Alerts staff via notifications if urgent action is needed.

- This software interacts with the multi-modal database to fetch and update patient data dynamically.

Lecture Schedule

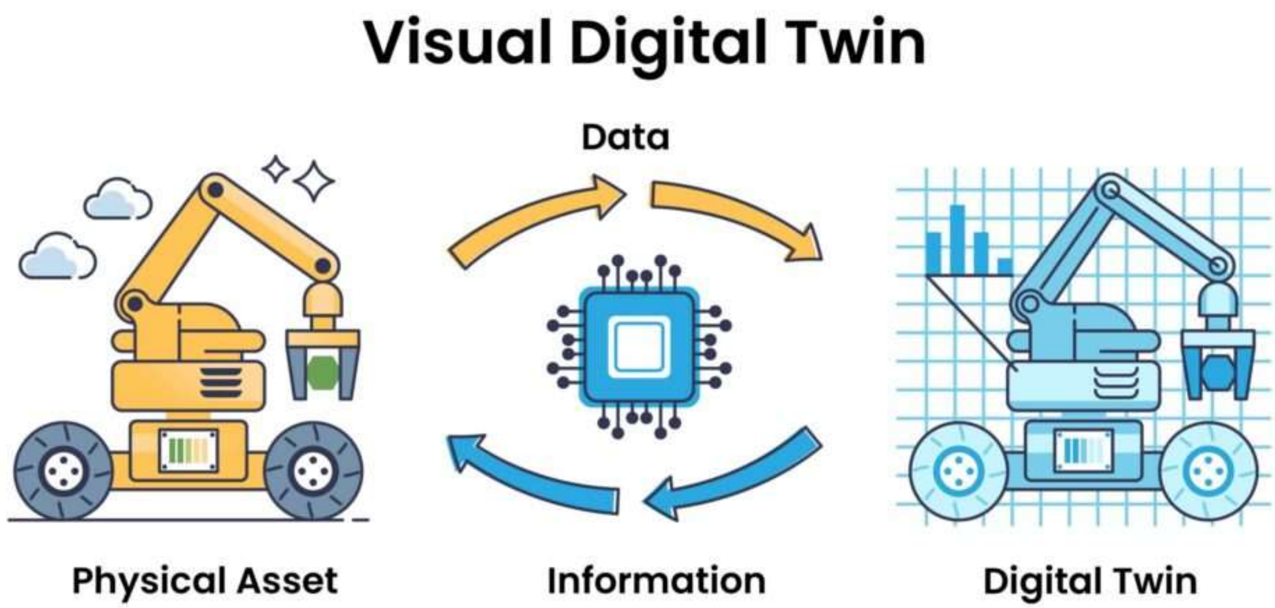
- Module 3 – Networking
  - IoT and Edge Computing





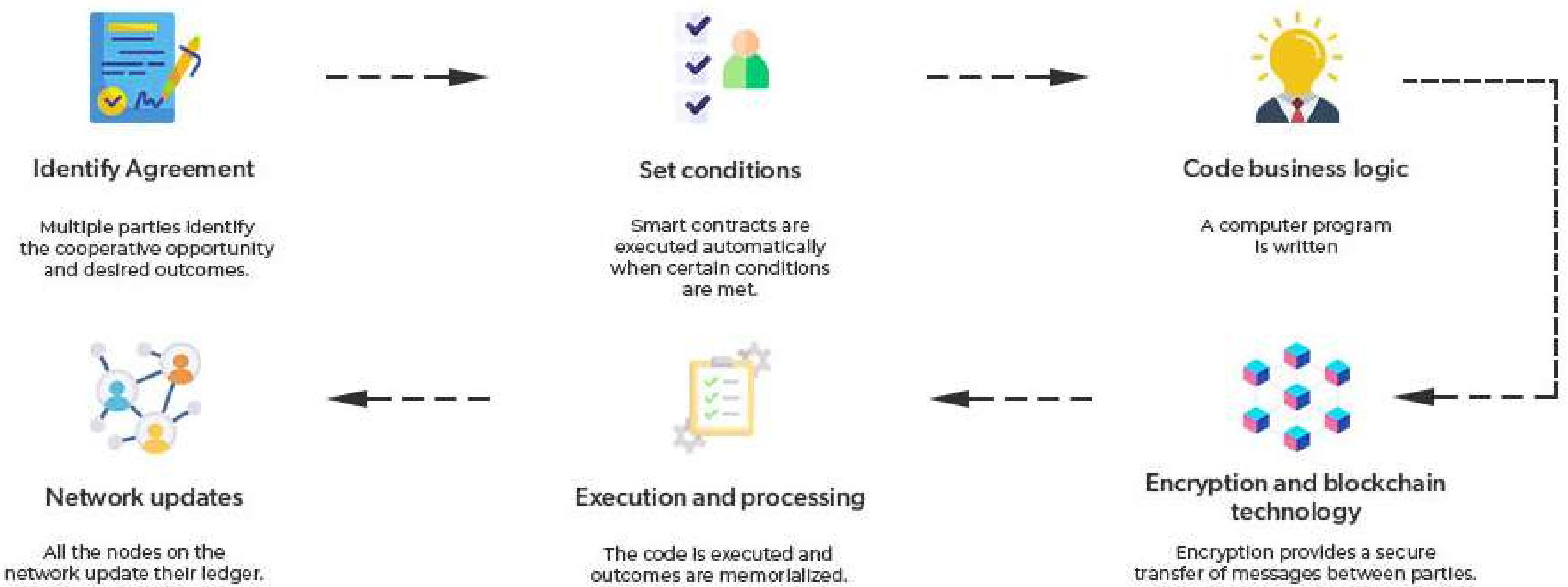
Lecture Schedule

- Module 3 – Networking
  - Digital twin technology



Lecture Schedule

- Module 3 – Networking
  - Blockchain and smart contract



Lecture Schedule

- Module 3 – Networking
  - Emerging cyber-attacks

Example

- For example, the system relies on a robust networking infrastructure:
  - Medical IoT devices send data over Wi-Fi/Bluetooth to central servers.
  - Secure connections (VPN, HTTPS) transmit data to cloud services.
  - Real-time communication between departments or even hospitals.

- Networking ensures continuous data flow between devices, databases, and software systems.

Advanced Database Overview

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Subject Topic Sequence

(1) Advanced Database Overview
(2) Spatial Data and Computing
(3) Graph Data and Computing
(4) Text Data and Processing

Multi-Modal Database

(1) AI4SE: artificial intelligence for software engineering
(2) Software Engineering in Cloud Computing I
(3) Software Engineering in Cloud Computing II

Software

(1) IoT and Edge Computing
(2) Digital Twin Technology
(3) Blockchains and Smart Contracts
(4) Emerging Cyber-Attacks

Networking

Outline

- Introduction to Database Systems
- Types of Databases
- Streaming Processing Systems
- Cloud-based Database Systems

Data

- Data is any distinct piece of information that is recorded or stored in a computer or other electronic device.

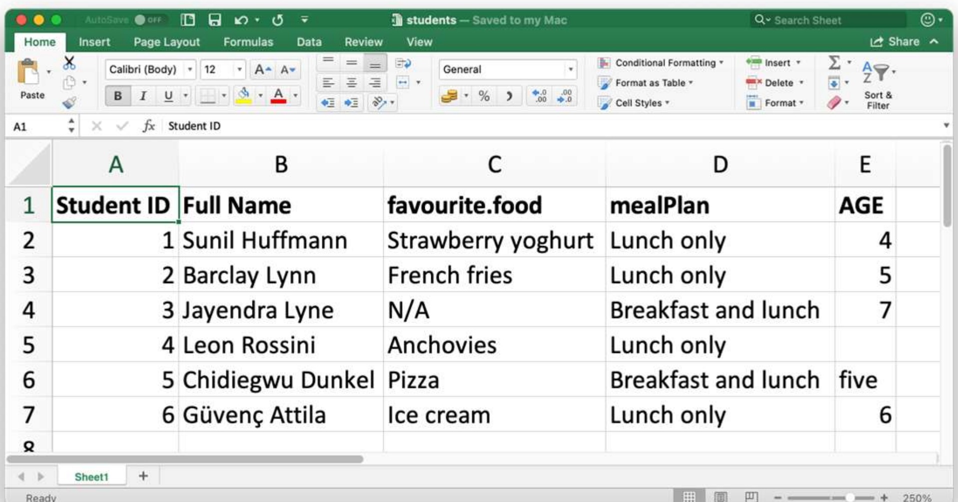
- It is a collection of unorganized facts, figures, and statistics that are meaningless until they are processed and turned into useful information.

- Data can be in different forms, including text, numbers, images, audio, and video.

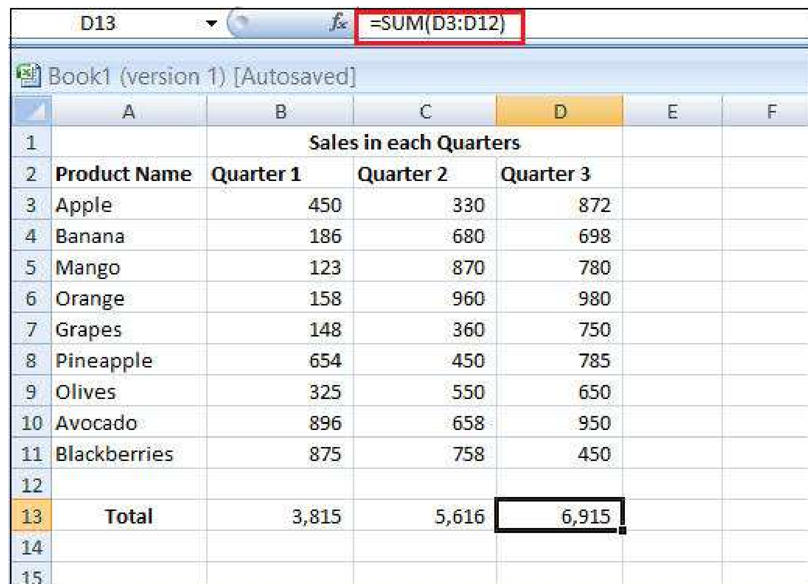
- Data is essential for decision-making, research, analysis, and problem-solving in many fields, including business, science, medicine, education, and government.

Data

- There are two main types of data:
  - **Structured data** is organized in a specific format, such as a spreadsheet or database, and can be easily processed and analyzed.



Student ID	Full Name	favourite.food	mealPlan	AGE
1	Sunil Huffmann	Strawberry yoghurt	Lunch only	4
2	Barclay Lynn	French fries	Lunch only	5
3	Jayendra Lyne	N/A	Breakfast and lunch	7
4	Leon Rossini	Anchovies	Lunch only	
5	Chidiegwu Dunkel	Pizza	Breakfast and lunch	five
6	Güvenç Attila	Ice cream	Lunch only	6



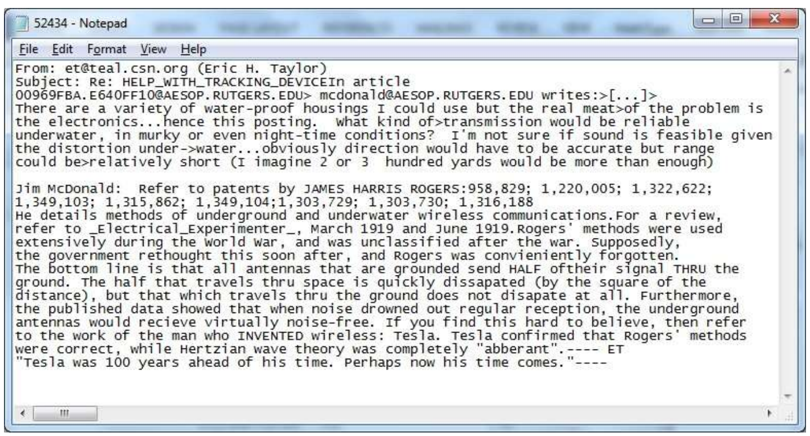
D13 =SUM(D3:D12)

Book1 (version 1) [Autosaved]

		Sales in each Quarters	
Product Name	Quarter 1	Quarter 2	Quarter 3
Apple	450	330	872
Banana	186	680	698
Mango	123	870	780
Orange	158	960	980
Grapes	148	360	750
Pineapple	654	450	785
Olives	325	550	650
Avocado	896	658	950
Blackberries	875	758	450
Total	3,815	5,616	6,915

Data

- There are two main types of data:
  - **Unstructured data**, on the other hand, is not organized in a specific format and may include text, images, or multimedia. It is more difficult to process and analyze unstructured data, but it can provide valuable insights and information.



Data

- Data can also be classified based on its quality.
  - **High-quality** data is accurate, complete, relevant, and timely. It is free from errors, inconsistencies, and duplications.
  - **Low-quality data** may be incomplete, outdated, or contain errors, which can lead to incorrect conclusions or decisions.

A customer's contact record includes:

- Full name: *Jane Smith*
- Email: *jane.smith@example.com*
- Phone number: *+61 412 345 678*
- Address: *123 Market Street, Sydney, NSW 2000*
- Last updated: *2 days ago*

Another customer's record includes:

- Name: *J. Smi*
- Email: *jane@.com*
- Phone number: *missing*
- Address: *Old info*
- Last updated: *5 years ago*

Data

- **Big data**: large datasets that are too complex or unstructured to be processed using traditional methods.

- Big data requires specialized tools and techniques, such as data mining, machine learning, and artificial intelligence, to extract insights and patterns.

Database

- A database is an organized collection of structured data that is stored and managed in a computer or other electronic device.
- A database typically consists of one or more tables that are related to each other.
  - Each **table** contains a set of records, and each record represents a specific instance of the data that is being stored.
  - The **columns** in a table represent different **attributes** of the data, such as name, age, or address. Each column is given a data type that specifies the type of data that can be stored in it, such as text, number, or date

Relational Database

- Relational databases organizes data into tables with predefined relationships between them.

- Relational databases use a standardized language called Structured Query Language (SQL) to manipulate data.
  - SQL is a powerful language that allows users to create, modify, and query databases.
  - It can be used to perform a wide range of operations, from simple queries to complex data transformations.

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Example

- A Customer Table

CustomerID	FullName	Email	Phone	Address	LastUpdated
1	Jane Smith	jane.smith@example.com	+61 412 345 678	123 Market Street, Sydney, NSW 2000	2025-08-03
2	Mark Lee	mark.lee@example.com	+61 412 987 654	456 King Street, Melbourne, VIC 3000	2025-08-01
3	Emily Chen	emily.chen@example.co m	+61 400 111 222	789 Queen Road, Brisbane, QLD 4000	2025-08-02

sql

```sql
SELECT FullName, Email
FROM Customers;
```

Expected Output:

FullName	Email
Jane Smith	jane.smith@example.com
Mark Lee	mark.lee@example.com
Emily Chen	emily.chen@example.com

Non-Relational Database

- Non-relational databases organize data in other ways such as key-value pairs or documents.

- Non-relational databases use a variety of languages and data models to manage data.

- Some of the most popular non-relational databases include MongoDB, Cassandra, and Redis.

Example

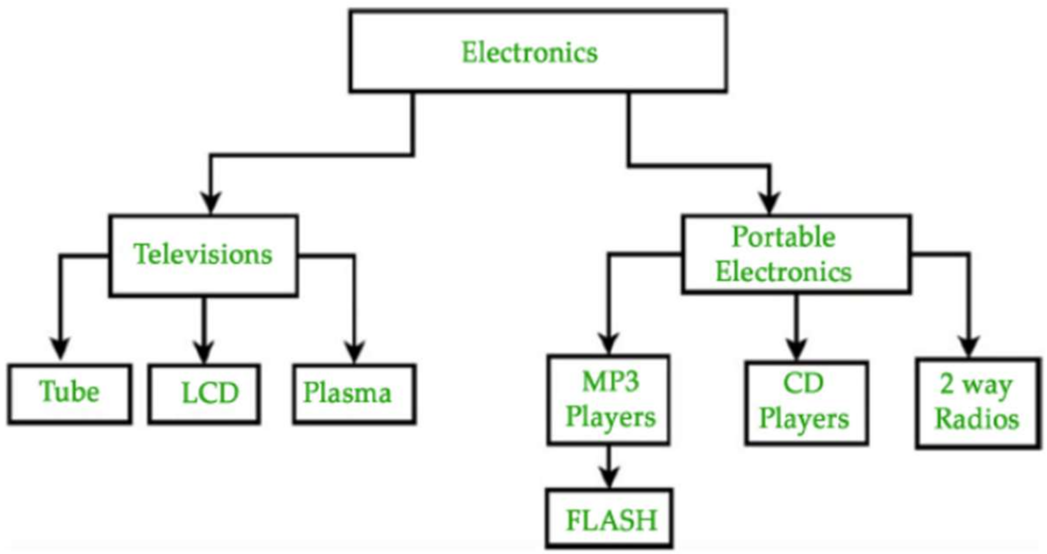
- Storing Customer Records in MongoDB

Each entry in the collection is a **document** (similar to a row in SQL, but more flexible).

```json
{
  "_id": "cust001",
  "fullName": "Jane Smith",
  "email": "jane.smith@example.com",
  "phone": "+61 412 345 678",
  "address": {
    "street": "123 Market Street",
    "city": "Sydney",
    "state": "NSW",
    "postcode": "2000"
  },
  "lastUpdated": "2025-08-03T10:00:00Z"
}
```

Hierarchical Database

- Hierarchical Databases are a type of database management system (DBMS) that organizes data in a tree-like structure, with each record having one parent and multiple children.



An example of the hierarchical database.

Example

- File System Structure

Root folder

- Folder: `/Documents`
  - File: `resume.docx`
  - Folder: `/Work`
    - File: `report.pdf`
- Folder: `/Pictures`
  - File: `vacation.jpg`

- Geographical Hierarchy

- Country: Australia
  - State: New South Wales
    - City: Wollongong
      - Suburb: Keiraville

Example

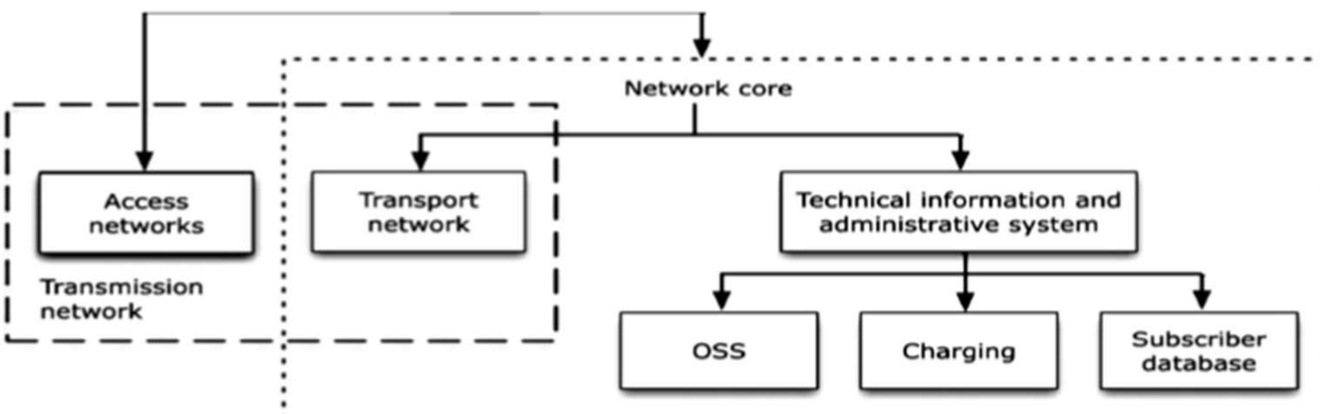
- XML / JSON Data

```xml
<company>
  <department name="Engineering">
    <employee id="1">Alice</employee>
    <employee id="2">Bob</employee>
  </department>
</company>
```

```json
{
  "company": {
    "department": {
      "name": "Engineering",
      "employees": [
        {"id": 1, "name": "Alice"},
        {"id": 2, "name": "Bob"}
      ]
    }
  }
}
```

Hierarchical Database

- Application
  - Used in telecommunications systems to manage customer data, billing information, and call records.
  - The efficient and performant nature of hierarchical databases makes it easy to handle the large volume of structured data that is typically found in telecommunications systems.



Hierarchy of networks and systems in a generic telecom infrastructure.

Hierarchical Database

- Advantage
  - They are well-suited for managing large amounts of structured data that has a predictable and fixed structure.
    - This makes them ideal for use in applications such as banking, insurance, and manufacturing, where there is a need to store and manage vast amounts of data with a consistent and hierarchical structure.

Hierarchical Database

- Advantage
  - They are highly efficient and performant. Since data is organized in a tree-like structure with a single root node at the top and multiple child nodes branching out from it, it is easy to navigate and access data quickly.
    - This makes hierarchical databases well-suited for applications where speed and performance are critical.
    - Hierarchical databases also offer a high degree of data security and control.

Hierarchical Database

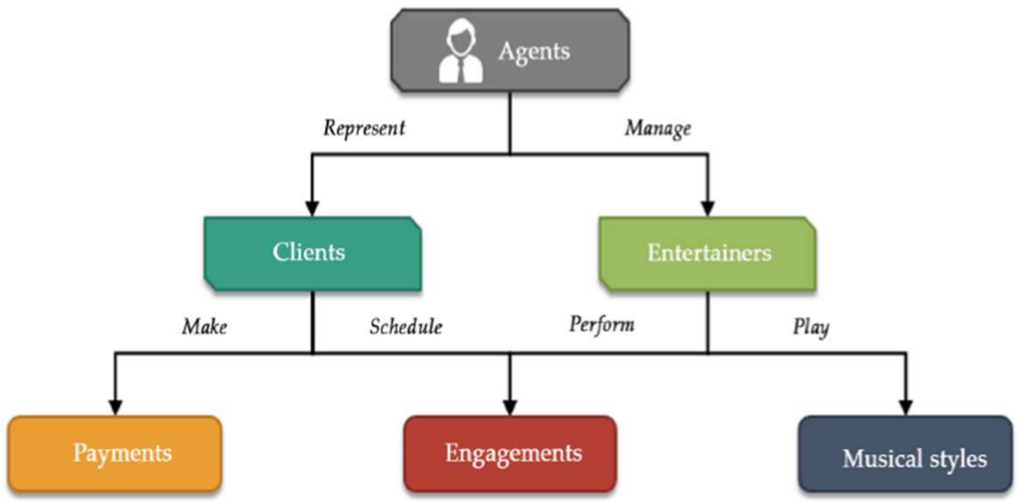
- Disadvantage
  - They are not well-suited for managing complex or unstructured data, and they can be difficult to modify or update when the underlying data structure changes.
  - This is why they have been largely replaced by more flexible and scalable database models, such as relational databases.

Network Database

- A network database is a type of database model that is designed to manage complex data structures and relationships between data entities.
- In a network database, data is organized into **records** and **sets**, with each set containing one or more records.
- The relationships between sets are defined by a network structure, which allows for many-to-many relationships between data entities.

Network Database

- Data is organized into a series of interconnected records, with each record containing one or more data fields. The records are organized into sets, with each set representing a different type of entity or object.



An example of the network database.

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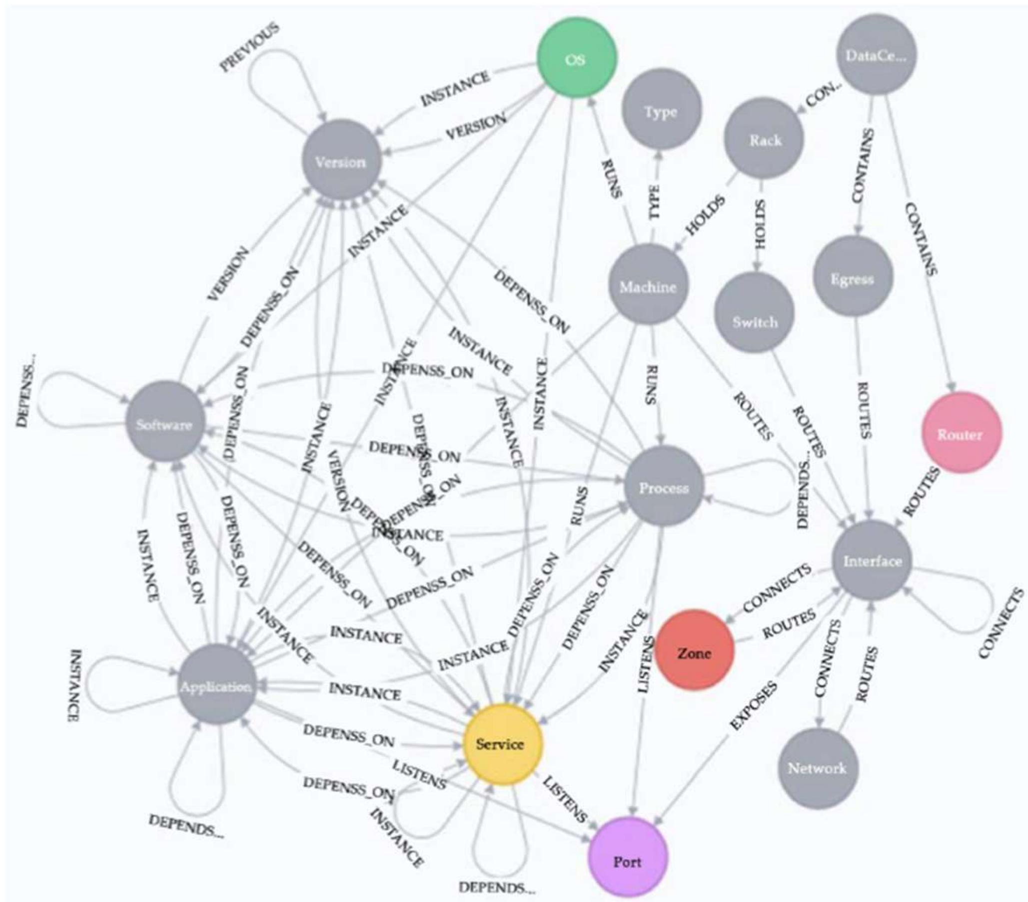
Network Database

- Application
  - In a database for a university, there might be sets for students, professors, courses, and departments.
  - Each set would contain records with data fields for specific attributes or properties of that entity.
  - The network structure in a network database is defined by a set of **pointers** that connect related records together.
  - Each record has one or more pointers that define its relationships to other records in the database.

Network Database

- Advantage
  - Ability to manage complex data relationships.
  - They are well-suited to applications where there are many-to-many relationships between data entities, or where the data structure is subject to frequent changes.
  - Network databases are also highly efficient in terms of data retrieval, since they use pointers to quickly navigate between related records.

Network Database



An example of the complex data relationship.

Network Database

- Disadvantage
  - They can be complex and difficult to design and maintain, and they are not as widely used as other database models such as the relational database model.
  - Additionally, they are not as scalable as other database models, since the network structure can become unwieldy as the database grows in size.

NoSQL Database

- NoSQL databases are databases that do not use the traditional structured query language (SQL) used in relational databases.
- They are designed to **handle large volumes of unstructured or semi-structured data**, which are increasingly becoming more common in modern applications.

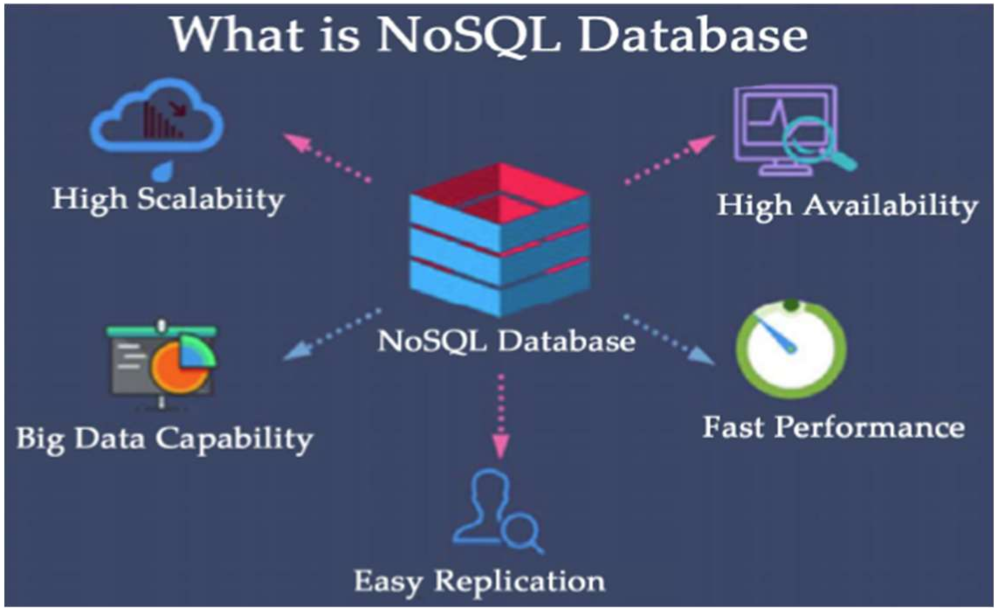


Illustration of the NoSQL database.

NoSQL Database

- This data can include text, images, videos, and other types of unstructured data, as well as data from distributed systems, social networks, and big data analytics.

- Unlike traditional relational databases, NoSQL databases use a variety of data models, including key-value stores, document-oriented databases, graph databases, and column-family stores, among others.

Example

- Key-value stores, for example, are simple databases that store data as key-value pairs, making them ideal for caching and other performance optimizations.

Key	Value
`cart:user123`	`{"items": ["book", "pen", "notebook"], "total": 25.50}`
`cart:user456`	`{"items": ["laptop", "mouse"], "total": 799.00}`
`cart:user789`	`{"items": [], "total": 0.00}`

Example

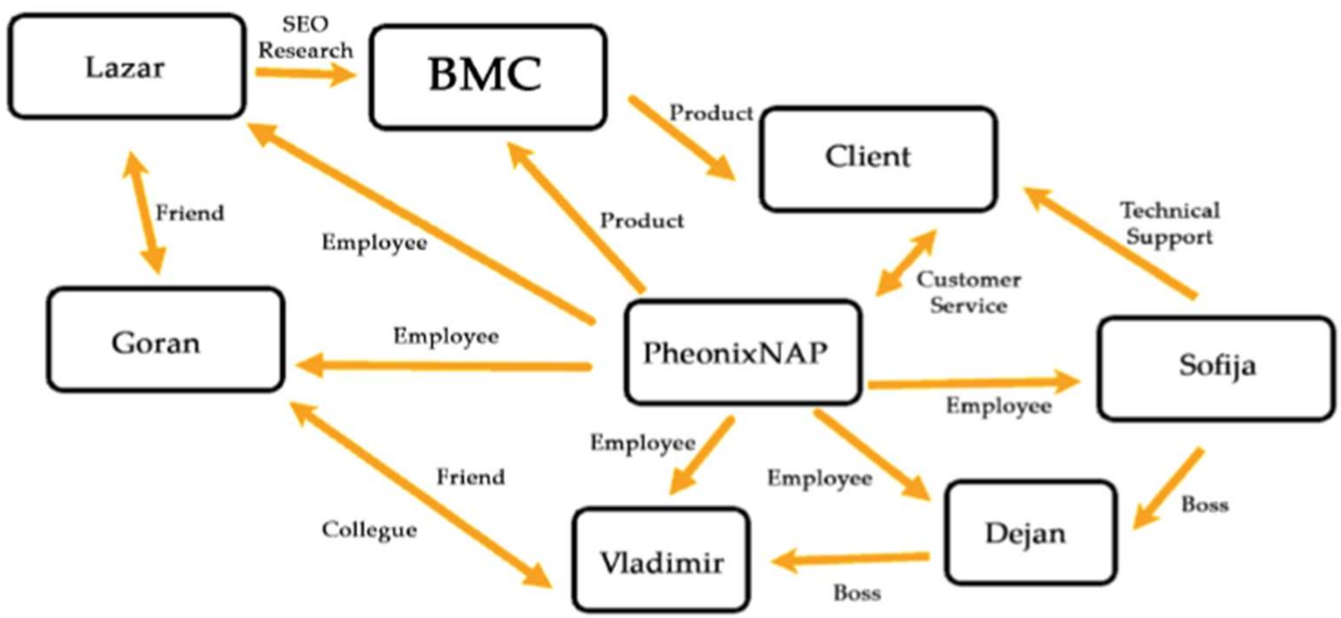


Illustration of an example of the NoSQL database.

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NoSQL Database

- Advantage
  - High scalability and flexibility
    - They can handle large volumes of data and distribute data across multiple servers, making them ideal for applications that require high availability and horizontal scalability.
    - NoSQL databases are also highly performant, as they can perform queries and updates quickly, without the overhead of traditional relational databases.
- Disadvantage
  - A lack of standardization and a higher learning curve for developers who are used to traditional relational databases.

Graph Database

- A graph database is a type of NoSQL database that uses graph theory to store, map, and query relationships between data elements.
- In a graph database, data is represented as nodes, edges, and properties, which can be used to model complex relationships between entities

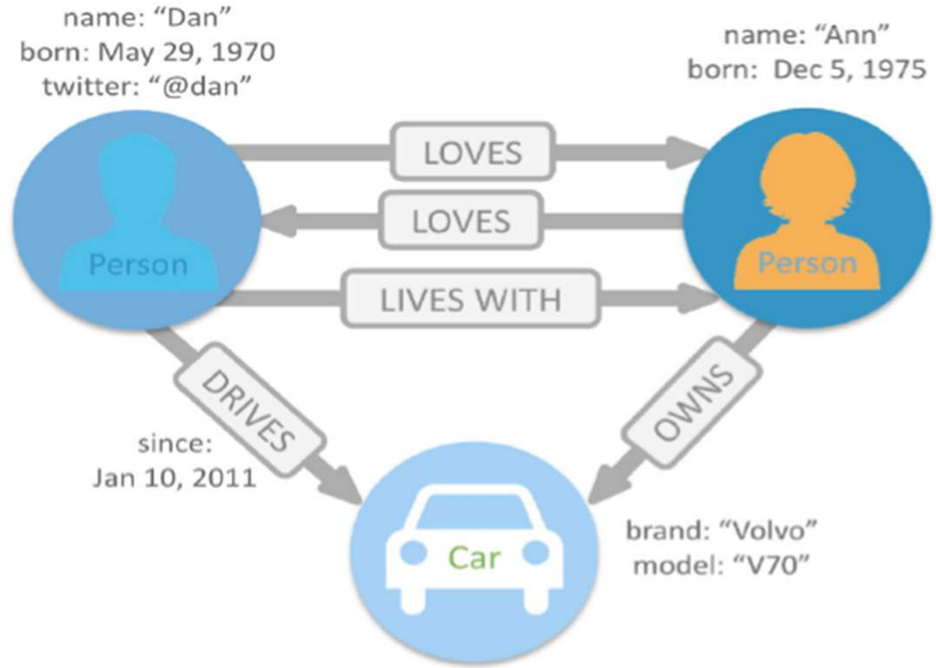


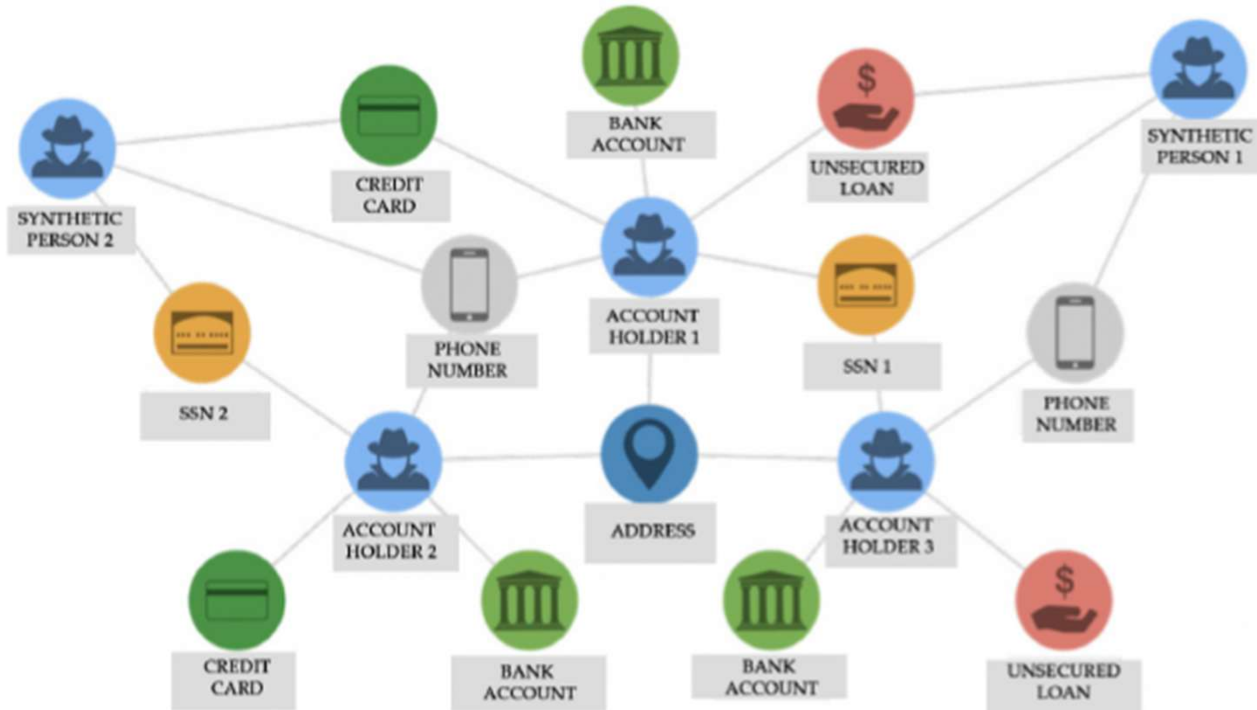
Illustration of an example of the NoSQL database.

Graph Database

- Nodes represent the entities in the database, while edges represent the relationships between those entities.
  - For example, in a social network graph database, a node might represent a person, while an edge might represent a connection between two people, such as a friendship or a follow relationship.
  - Properties provide additional information about the nodes and edges, such as a person’s name or a connection’s strength.

Graph Database

- Application



Fraud detection using graph database

Graph Database

- Advantage
  - Flexibility
    - Graph databases are highly flexible and can be used to model a wide variety of data structures, including complex relationships between entities. This makes them ideal for use in applications where data is constantly changing or evolving.
  - Performance
    - Graph databases are designed to be highly performant, even when dealing with large and complex datasets. Because graph databases use indexing and caching techniques to retrieve data quickly, and they are optimized for querying relationships between entities.
  - Scalability
    - Graph databases are highly scalable, meaning that they can handle large amounts of data and users without sacrificing performance. This makes them ideal for use in applications with high growth potential.

Graph Database

- Advantage
  - Schema-Less Design
    - Graph databases do not require a fixed schema, meaning the database can be easily adapted to changing business without requiring significant modifications to the underlying model.
  - Real-Time Analysis
    - Graph databases can be used to perform real-time analysis of data, making them ideal for use in applications where quick decision-making is required.
  - Natural Language Processing
    - Graph databases are well-suited for natural language processing applications, where relationships between entities and need to be identified and analyzed.

Graph Database

- Disadvantage
  - Complexity
    - Graph databases can be more complex to implement and maintain than other types of databases. This is because they require a deeper understanding of graph theory and specialized query languages, and may require more advanced technical expertise to manage.
  - Data Modeling Challenges
    - Graph databases require careful consideration of data modeling to properly represent relationships between entities. This can be a challenge, especially when working with complex data structures or large datasets.

Graph Database

- Disadvantage
  - Cost
    - Some graph database solutions may be more expensive than other types of databases, which can be a barrier to adoption for smaller organizations or projects.
  - Performance Limitations
    - While graph databases are designed to be highly performant, their performance may be limited in some use cases. For example, when working with large or complex graphs, or when performing complex queries that require traversing multiple relationships.

Document Database

- Document databases are a type of NoSQL database that are specifically designed to store and manage semi-structured and unstructured data.

- Unlike traditional relational databases, which rely on a rigid schema to organize data, document databases allow for flexible and dynamic data modeling.

- In document databases, data is stored as JavaScript Object Notation (JSON) documents. These documents can be nested and can include arrays and other complex data structures. This allows for more natural and intuitive data modeling, since the structure of the data can be tailored to the specific needs of the application.

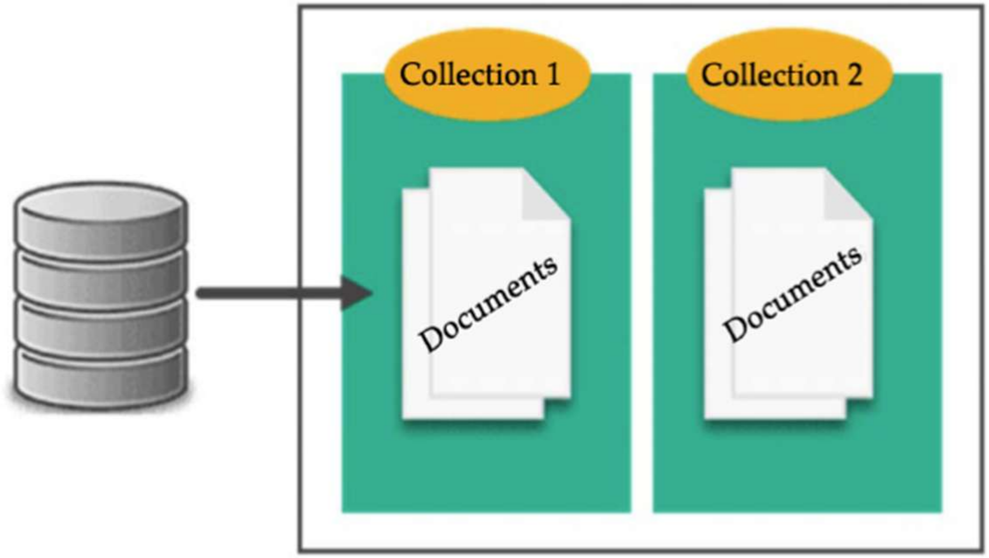


Illustration of the document database.

Document Database

- Application

Illustration of the content management system.

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Document Database

- Advantage
  - Flexible Data Modeling
    - Document databases allow flexible data modeling, enabling businesses to store complex and unstructured data in a structured format. This makes it easier to manage and process data and to scale the database system as the business grows.
  - Scalability
    - Document databases are highly scalable, both horizontally and vertically.
    - Horizontal scalability is achieved by distributing data across multiple servers, while vertical scalability involves increasing the single server.
  - High Performance
    - Document databases can handle large amounts of data and complex queries quickly and efficiently. They use advanced indexing techniques to improve query performance and support fast read and write operations.
  - Easy to Use
    - Document databases are often easier to use than traditional relational databases. They do not require a predefined schema, so developers can start building applications immediately without having to design a schema.

Document Database

- Advantage
  - Developer Productivity
    - Document databases allow developers to work with data in the same format as their programming language, reducing the need for data mapping and conversion. This increases developer productivity and reduces development time.
  - Cost-Effective
    - Document databases can be less expensive to run than traditional databases. They require less hardware, and the open-source versions are often free to use.
  - Availability
    - Document databases are highly available, with many offering automatic failover and replication capabilities. This ensures that data is always accessible, even in the event of a server failure.
  - Schema Evolution
    - Document databases allow for schema evolution, which means that the database schema can evolve over time as the data changes. This reduces the need for downtime or complex schema migrations.

Document Database

- Disadvantage
  - Lack of Standardized Query Language
    - Unlike relational databases, document databases lack a standardized query language. This can make it difficult for users to interact with the database, a specialized as they must learn a new query interface to interact with the data.
  - Limited Scalability
    - Document databases are generally less scalable than other types of databases, which can limit their use in large-scale applications. This is because document databases often store data in a single machine or cluster, making it difficult to scale horizontally.
  - Limited Support for Transactions
    - Many document databases lack support for transactions, making it difficult to maintain data integrity and consistency. This can be particularly problematic for applications that require ACID compliance.
  - Limited Support for Analytics
    - Document databases are often less suitable for data analysis and reporting than relational databases. This is because document databases lack the structured data necessary for traditional analytics tools and reporting systems.
  - Data Duplication
    - Document databases often duplicate data across multiple documents, which can lead to redundancy and wasted storage space. This can also make it more difficult to maintain data consistency and integrity.

Streaming Processing Systems

- Stream processing systems are designed to handle continuous and real-time data streams that are generated at high velocity. They can process data as it is generated, without waiting for the data to be collected and stored in a database.
- This makes stream processing systems ideal for applications that require real-time data analysis and decision-making.

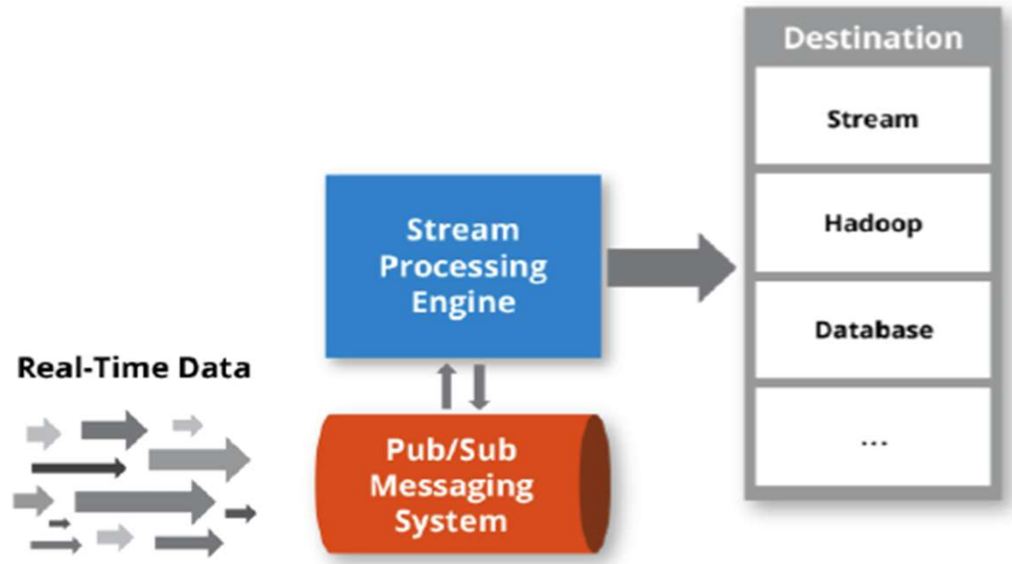


Illustration of the continuous and real-time data processing.

Streaming Processing Systems

- Stream processing systems typically use in-memory processing techniques to enable high-speed data processing.

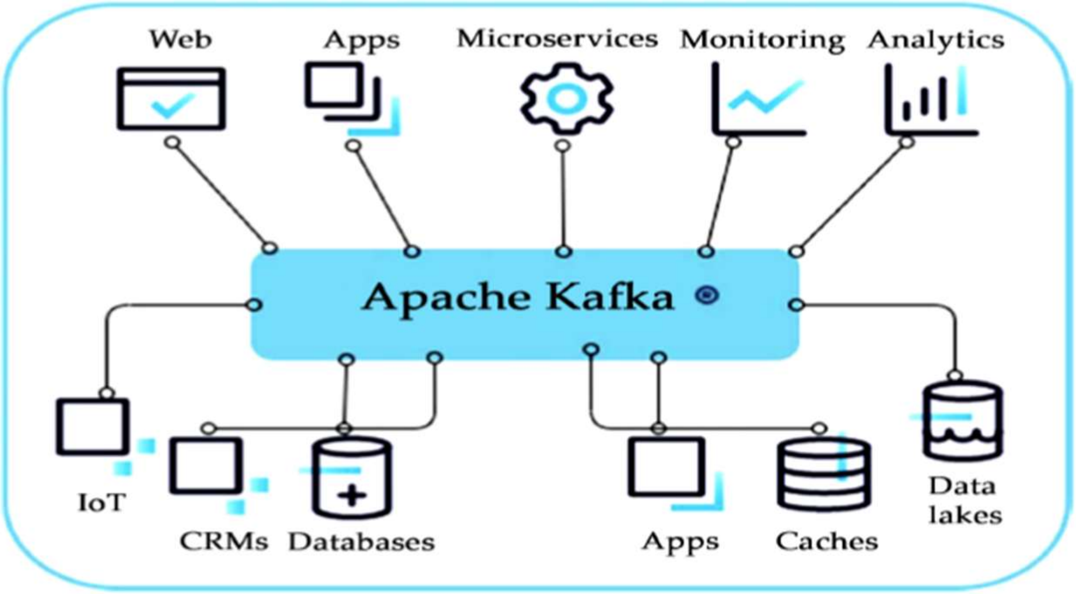
- Stream processing systems are typically designed as distributed systems that can process data across multiple nodes in a cluster.
  - Distributed processing enables stream processing systems to scale horizontally and handle large volumes of data.
  - It also provides fault tolerance, as data can be replicated across multiple nodes in the cluster, ensuing that data is not lost if a node fails.

Streaming Processing Systems

- Stream processing systems are designed to be highly scalable and fault-tolerant.
  - They can handle large volumes of data and can be scaled up or down as needed.
  - They are also designed to handle node failures gracefully, by replicating data across multiple nodes in the cluster.
  - This ensures that data is not lost if a node fails and that the system can continue to operate smoothly.
  - Scalability and fault tolerance are critical characteristics of stream processing systems , as they enable these systems to process data reliably at scale

Example

- Apache Kafka is an open-source distributed streaming platform that is used for building real-time data pipelines and streaming applications.
- Kafka is designed to handle high-throughput, low-latency data streams and provides reliable, fault-tolerant data delivery.
- It is widely used for applications such as data ingestion, real-time analytics, and data processing.



Schematic of the Apache Kafka

Streaming Processing Systems

- Application
  - **Real-time analytics** is a common application of stream processing systems.
    - Stream processing systems enable organizations to perform real-time analysis on streaming data, which can provide immediate insights and enable rapid decision-making. Real-time analytics applications include financial trading, online advertising, and supply chain management.
  - **Fraud Detection Stream processing systems** are also used for fraud detection applications.
    - By analyzing real-time data streams, stream processing systems can detect fraudulent transactions or activities in real-time. Fraud detection applications include credit card fraud detection, insurance fraud detection, and network security.
  - **Internet of Things (IoT)**
    - Stream processing systems are an important tool for processing and analyzing data from IoT devices. By processing and analyzing data in real-time, stream processing systems can enable real-time decision-making and provide valuable insights into the behavior of IoT devices. IoT applications include smart homes, industrial automation, and smart cities.

Streaming Processing Systems

- Application
  - **Social Media Analysis**
    - By processing social media data streams, stream processing systems can provide valuable insights into social media trends, sentiment analysis, and customer behavior.
    - Social media analysis applications include marketing analytics, reputation management, and social media monitoring. Each of these stream processing applications provides unique insights and value to organizations that leverage real-time data processing.
    - By using stream processing systems , organizations can enable real-time decision-making, gain competitive advantage, and provide superior customer experiences.

Streaming Processing Systems

- Challenge
  - Scalability and Performance
    - As data volumes continue to grow, stream processing systems must be able to scale horizontally across multiple nodes in a cluster. Stream processing systems must also provide high-performance processing capabilities to enable real-time analytics and decision-making.
  - Data Consistency and Data Quality
    - Stream processing systems must ensure that data is consistent and accurate across multiple nodes in the cluster, and that the data is of high quality. This requires robust data validation and cleansing techniques to ensure that the data is reliable and trustworthy.
  - Integration with Machine Learning
    - By integrating with machine learning techniques, stream processing systems can enable advanced analytics and predictive modeling on real-time data streams. This will enable organizations to gain deeper insights into their data and make more informed decisions

Cloud-based Database Systems

- Assume that user connects to cloud database via internet and computer. The data centers, cloud data centers, and the user viewing the data are all linked by the internet, which serves as a bridge between all of these organizations.
- For this application, peer-to-peer communication is preferred. Because a single node may process every user-implemented query, peer-to-peer communication is used.
- Despite its appearance, this type of node system has a straightforward solution: Each cloud database node has a map of the data it stores. The data for the specific query is easier to retrieve thanks to this map to the stored data.

Cloud-based Database Systems

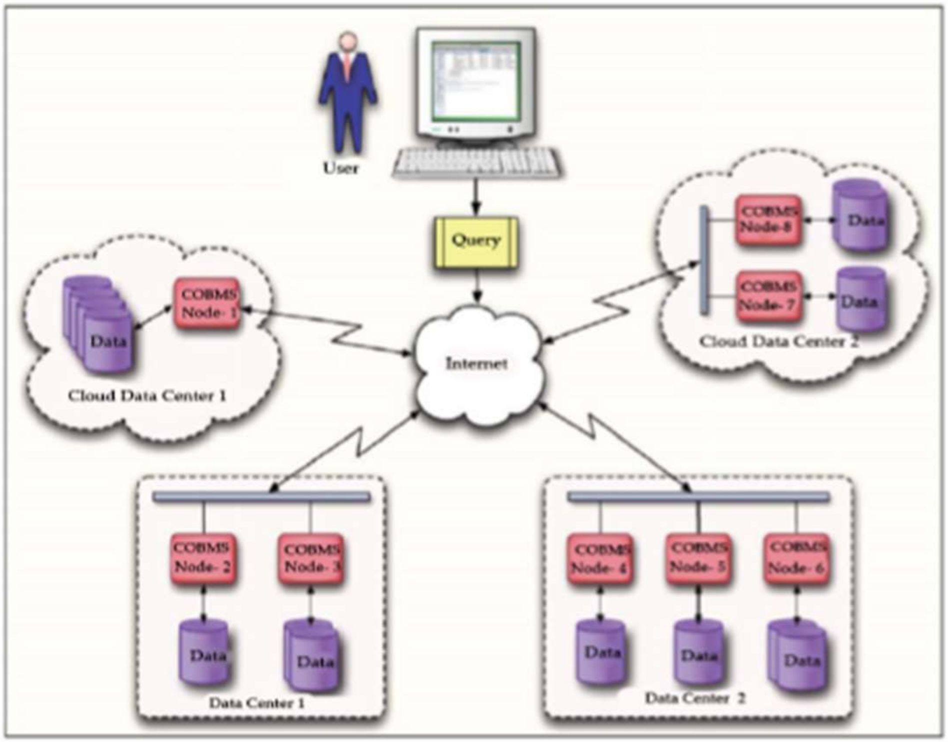


Illustration of the structure of cloud.

Cloud-based Database Systems

- Challenges
  - Internet Speed
    - The data transfer speed is relatively very quick when compared to internet speed utilized to connect to data center. Performance of cloud database is hampered by this.
    - This has an effect on how well cloud database functions. How quickly queries are sent to database and data is received from the data center depend on the internet's speed.
    - The solution to this issue is faster speed lines, however these are extremely expensive and defeat the purpose of using a cloud database.

Cloud-based Database Systems

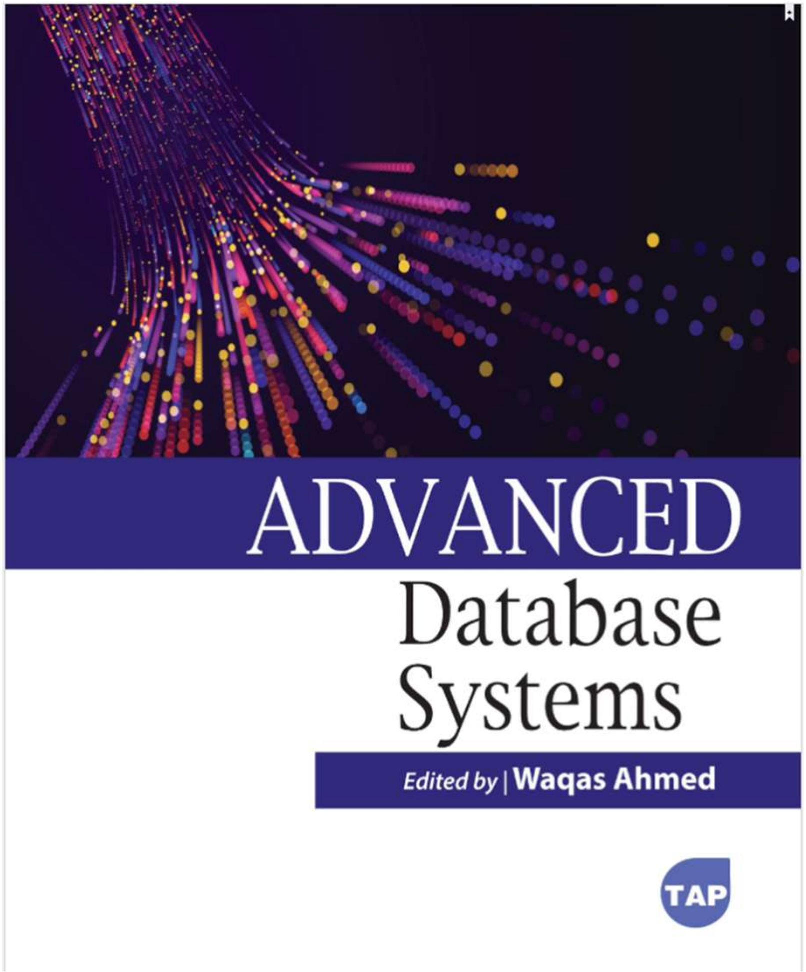
- Challenge
  - Elastic Scalability
    - When we're talking about cloud databases, any workload can be managed by a good cloud database. Yet, the problem in a cloud database arises when workload exceeds system capacity. Cloud database must be able to scale out on its own as workload grows. The database's scaling out helps cloud database achieve its best performance and efficiency.
  - Privacy has been the primary issue with cloud computing.
    - Cloud computing is more advanced in terms of accessibility to consumers as well as hackers who desire to break into the system. It's critical that the cloud database, which houses client records for organizations, is secure. The companies cannot risk having their database's data leak. If the database has data encryption, securely storing data is relatively easy to do.

Cloud-based Database Systems

- Challenge
  - Multi-Tenancy
    - The devices on which the workloads are distributed and the joining method used can be identical. The ideal solution is to create virtual machines for each database and to build several virtual machines on the same system. Additional equipment is required. Assume that the same task needs to be split between two or three machines. The performance and speed ultimately suffer a 6– 10-fold decline as a result.
    - Each virtual machine has its own operating system and database, which results in the decreased performance. When these two essential components are split, each virtual machine has its own buffer loop. It would be desirable to use the same database server across numerous machines since this would increase performance.

Reference

- **Advanced Database Systems**, edited by Waqas Ahmed, Toronto Academic Press, 2024. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uow/detail.action?docID=31545566.



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